

Product Carbon Footprint Study Report

Product Name: fan, for power supply unit, desktop computer

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: fan, for power supply unit, desktop computer

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1kg fan, for power supply unit, desktop computer

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg fan, for power supply unit, desktop computer.

2. System Boundary

fan production, for power supply unit, desktop computer, in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of fan, for power supply unit, desktop computer produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **11.791 kg CO2-Eq**.

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

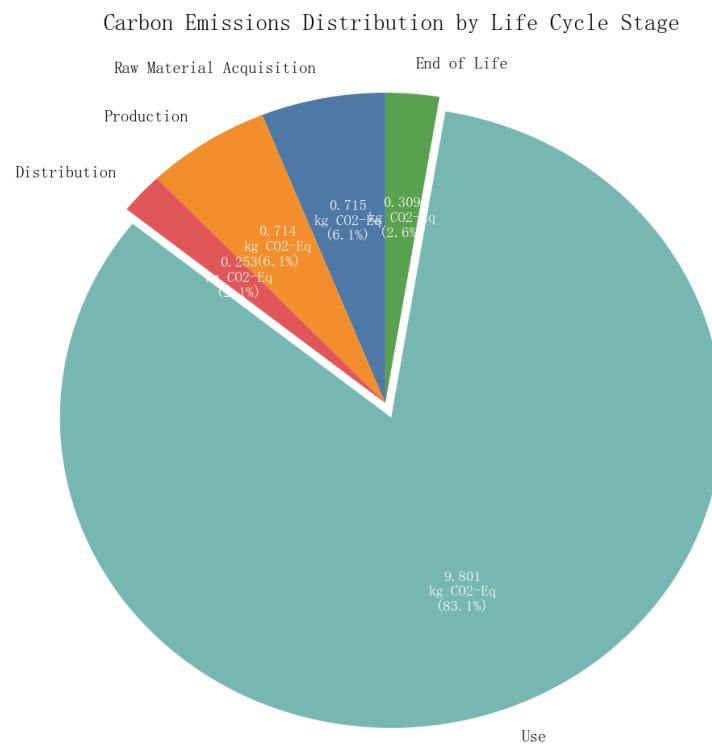


Table 1: Carbon Emissions by Life Cycle Stage

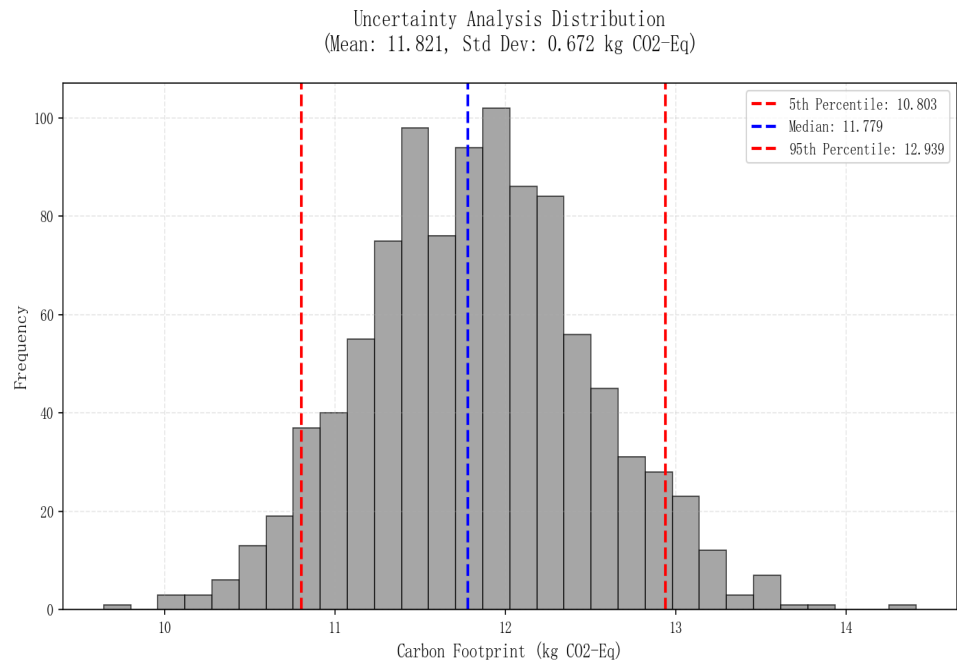
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	0.715	6.06
Production	0.714	6.05
Distribution	0.253	2.15

Use	9.801	83.12
End of Life	0.309	2.62
Total	11.791	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO2-Eq)
Mean	11.821
Standard Deviation	0.672
Coefficient of Variation (CV)	5.69%
Median	11.779
5th Percentile	10.803
95th Percentile	12.939

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the fan for the power supply unit in a desktop computer is 11.791 kg CO₂-Eq, which is moderate for a small electronic component but significant when aggregated across millions of units. The lifecycle contribution analysis reveals a highly skewed environmental profile: the use phase dominates at 83.1% (9.801 kg CO₂-Eq), while raw material extraction and production contribute only 6.1% each, and distribution and end-of-life are negligible at 2.1% and 2.6%, respectively. This indicates that the fan's operational electricity consumption is the overwhelming source of emissions, dwarfing upstream manufacturing and downstream disposal impacts. Consequently, the major emission source is the electricity required to power the fan over its service life, with the specific emission factor of the regional grid playing a pivotal role. Strategies to reduce the total footprint must therefore prioritize improvements in energy efficiency and the use of low-carbon electricity rather than focusing solely on material or production changes.

2. Statement of Assumptions and Limitations

Key assumptions underpinning this study include a typical usage scenario for a desktop computer fan—likely 8 hours/day over a 3–5 year lifespan—and an average grid electricity mix, e.g., global or regional default. The use phase calculation depends critically on these behavioral and energy-source assumptions; any deviation (e.g., longer daily operation or higher carbon grid intensity) would substantially alter the results. Data limitations are evident: the top process contributions are all listed as “Unknown,” which reveals a lack of process-level allocation detail in the LCA model. This gap reduces the ability to pinpoint specific manufacturing or supply chain hotspots. Additionally, the dataset may rely on generic industry averages rather than supplier-specific primary data, introducing uncertainty that is partially captured by the coefficient of variation (5.69%) and the 90% confidence interval (10.803–12.939 kg CO₂-Eq). System boundaries are apparently cradle-to-grave, but excludes capital goods, infrastructure, and potential secondary use scenarios.

3. Suggestions for Improvement

- ****Improve energy efficiency during use:**** Redesign the fan to reduce power consumption (e.g., using brushless DC motors, optimized blade aerodynamics, or variable-speed control). Target a 20% reduction in wattage to cut use-phase emissions by an equivalent proportion.
- ****Promote low-carbon electricity:**** Encourage consumers and data centers to power computers with renewable energy. For manufacturers, offset use-phase emissions through certified renewable energy certificates (RECs) or power purchase agreements.
- ****Enhance material circularity:**** Substitute virgin plastics and metals with recycled content (e.g., post-consumer recycled ABS for the housing, recycled copper for windings) to lower raw material contributions by at least 2–3%.
- ****Extend product lifespan:**** Improve bearing reliability and motor durability to increase the fan's service life from 40,000 to 60,000 hours, thereby amortizing the production impact over more operational hours.

- ****Improve data transparency:**** Require suppliers to disclose process-level data (e.g., injection molding, motor winding, packaging) to replace the “Unknown” top processes, enabling more targeted improvement measures and reducing uncertainty.